

sionals in their industry. In the modern age, data scientists are playing a multi-faceted role of a computer professional, a mathematician, a statistician, and a scientist.

Educational Background

Individuals with a bachelor's degree in Mathematics, Economics, Computer Science, Social Sciences, Physical Sciences, and Statistics can opt for a career in Data Science. Study trends show that an acquired qualification plays a crucial role in the career of a data scientist. The maximum percentage of data scientists comes from Mathematics and Statistics (32%), followed by Computer Science (19%) and Engineering (16%) backgrounds. 88 percent data scientists possess a Master's degree and 46 percent have PhDs.

Technical Skills

Data scientist profiles ask for technical skills across multiple disciplines, domain knowledge, and best industry practices. These may include:

Programming languages and Tools

Data scientist profiles must have a fundamental understanding of multiple programming languages such as Python, Perl, Java, C/C++, and SQL. Individual capability to write, modify, and develop efficient code is of primary importance. One should have expertise in one of the programming languages. According to a survey, Python was the most preferred choice among programming languages (40% share) for the role of a data scientist.

1) R programming is one of the most preferred data analytical tools in Data Science, with up to 43 percent data scientists using this as a professional skill. R was built with the specific purpose of matching the complex needs of data science. Though the R learning curve is not simple, many aspirants are still learning it for data science to match the best industry standards.

2) Statistics and probability: Statistics and probability leverage the mathematical concepts in the learning curve for data science. Several combinations of basic and advanced features add more predictive



Statistics and Probability for Data Science

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